Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/12/2024
PI Name:	Simpson, Richard Ph.D.		
Project Title:	Promoting Behavioral Health, Cognitive, Sensorimotor and Immune Function Using Guided Imagery to Augment Exercise Training in an Isolated and Confined Spaceflight Analog Environment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0001	**Congressional District:**	3
Comments:	NOTE: Formerly at University of Houston until September 2017 move to University of Arizona.		
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	08/05/2019	**End Date:**	06/30/2025
No. of Post Docs:	3	**No. of PhD Degrees:**	3
No. of PhD Candidates:	5	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2025 per NSSC information (Ed., 5/17/24) NOTE: End date changed to 06/30/2024 per S. Mack/JSC (Ed., 7/11/23) NOTE: End date changed to 06/30/2023 per NSSC information (Ed., 3/3/23) NOTE: End date changed to 01/31/2023 per NSSC information (Ed., 3/10/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gordon, Judith Ph.D. (University of Arizona) Connaboy, Christopher Ph.D. (Rosland Franklin University)		
Grant/Contract No.:	80NSSC19K1480		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Operating in isolated, confined, and controlled (ICC) or extreme (ICE) environments impacts physical status, psychological symptoms, and cognitive function, all serving to degrade crew performance and jeopardize mission success. Countermeasures, such as exercise training help to reduce such risk, but individuals are still susceptible to problems adapting to and operating in ICC/ICE environments. Physical training has been shown to have positive benefits on cognitive and immune functions, and psychological status of individuals. However, importantly, increased stress levels have been shown to substantially limit individual engagement with these types of physical activity. Therefore, to ensure the efficacy of a physical training countermeasure, strategies need to be developed to ensure crewmembers maintain sufficient levels of training stimuli to provide these protective effects, while in the presence of the increased levels of stress associated with operating in ICC/ICE environments. The overarching hypothesis of this proposal is that structured exercise training performed in an isolated and confined spaceflight analog will ameliorate stress-induced changes in behavioral health, cognitive, sensorimotor, and immune system function. We further posit that adding a guided imagery (GI) intervention component to the exercise training regimen will increase exercise adherence and positive affective responses, and that the beneficial effects of exercise on behavioral health, cognitive, and immune system function will be mediated by reductions in stress and improvements/maintenance of physical health. The specific aims are as follows: SA1. Determine the effects of an in-flight validated exercise regimen (‘SPRINT’) on behavioral health, fatigue, cognitive, sensorimotor, and immune system function in an ICC spaceflight analog; SA2. Determine the impact of guided imagery on exercise adherence and positive affective responses to exercise in the ICC spaceflight analog. We will also explore the effects of adding guided imagery to the exercise regimen on behavioral health, fatigue, cognitive, sensorimotor, and immune system function; SA3. Determine if exercise training, with or without guided imagery, has a positive impact on biomarkers of stress and central nervous system function in the ICC spaceflight analog and can increase resilience during an acute stress model of sleep deprivation relative to the control group. We will use the Human Exploration Research Analog (HERA) and recruit crewmembers over 12 separate 45-day HERA missions. We anticipate enrolling participants from 3 campaigns (4 crewmembers per mission) over a 3-year period (n= 48). Four HERA missions (n=16) will receive the in-flight validated Integrated Aerobic and Resistance Exercise Training Prescription (‘SPRINT’) protocol only; Four HERA missions (n=16) will receive the same training regimen but with GI to augment the exercise response; And four HERA missions (n=16) will serve as the control group, receiving access to the exercise equipment but no GI or ‘SPRINT’ protocol. Measures of behavioral health, cognitive, sensorimotor, and immune (NK-cell cytotoxicity) function will be measured before, during, and after the mission, and the mediating/correlating effects of stress and central nervous system function will be determined by measuring biomarkers (cortisol, dehydroepiandrosterone (DHEA), Macrophage Inflammatory Protein (MIP), brain-derived neurotrophic factor (BDNF)) longitudinally in saliva. On completion of this project we expect to: (1) determine if the ‘SPRINT’ protocol is sufficient to maintain behavioral health of individuals during a spaceflight stress analog; and (2) ascertain whether or not the addition of a GI can augment the exercise training effects on ameliorating stress (including acute stress derived from sleep deprivation) and central nervous system dysfunction to maintain behavioral health, cognitive, sensorimotor, and immune system function in a high fidelity, isolated, confined, and controlled spaceflight simulation analog.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will increase understanding of how exercise training can offset the deleterious effects of isolation and confinement stress on immune system dysfunction and detriments in behavioral health. This has multiple applications to life on Earth, particularly in light of global lock-down procedures that were implemented during the COVID-19 pandemic.
	Cumulative Progress Report up to April 12th 2024 The "Promoting Health Through Exercise and Imagery" (PHETI) study was initiated in 2019 but experienced significant delays due to the COVID-19 pandemic. Whilst waiting on the NASA Human Exploration Research Analog (HERA) Campaign 6 to begin, the study team focused on identifying psychological 'targets' that could be manipulated through guided imagery to improve exercise adherence. This involved an extensive review of the literature which resulted in a completed MS thesis by student Jamie Lee-Elliot (advisors: Judith Gordon and Richard Simpson). This thesis is currently being re-formatted for publication in a scientific journal. A poster presentation was delivered at the NASA Human Research Program Investigators' Workshop (IWS) in Galveston, TX in February 2023 and in February 2024. HERA Campaign 6 Mission 1 began October 1, 2021. As of April 12, 2024, we have received full sample details from 16 participants across the 4 missions in Campaign 6. Mission 1 of Campaign 7 has been completed and we are awaiting sample delivery from NASA Johnson Space Center (NASA-JSC). The following analytical procedures have been completed for all 16 participants of Campaign 6: 1. Complete blood counts and immunophenotyping 2. NK-cell cytotoxicity assays in vitro 3. Aerobic fitness and isometric strength testing pre and post mission 4. Exercise adherence and exercise volume analysis during the mission 5. Serum biomarkers for stress and cognitive function, including cortisol, epinephrine, norepinephrine, DHEA-s and BDNF. Below is an abstract summary of our findings and conclusions to date. This will form the basis of an MS thesis defense for Ms. Elizabeth Beattie (scheduled for May 3, 2024). Abstract of Interim Campaign 6 data NASA considers spaceflight as an isolated, confined, and extreme environment (ICE), which consequently embodies multiple hazards that may impair both the physical and psychological well-being of the astronauts, implying that robust

Task Progress:	countermeasures to minimize health risks to the crew are vital. Immunity may be dysregulated due to various ICE factors, particularly chronic stress. This is a critical concern as it amplifies health risks, such as viral reactivation and impaired anti-tumor immune surveillance, especially in combination with other ICE aspects. An abundance of literature has shown that exercise has a mitigating effect on chronic stress and can also preserve and improve the normal functioning of the immune system, but it is not known if exercise performed in an ICE analog can impact potential changes in biomarkers of stress and immune function. The aim of this study was to determine the effects of a 45-day terrestrial Human Exploration Research Analog (HERA) mission on immune cell phenotypes and function (NK-cell cytotoxicity), and blood biomarkers of stress (cortisol, epinephrine, norepinephrine, DHEA-S) and cognitive function (BDNF). We also sought to determine voluntary exercise patterns among the crew and their associations with stress and immune biomarkers. Surprisingly, we found that biomarkers of stress and cognitive function did not change significantly in response to the HERA mission. There was, however, a decrease in NK-cell cytotoxicity during the late stage of the mission, which has also been documented previously in astronauts aboard the International Space Station. The crew were found to perform aerobic exercise at a consistent volume (intensity x duration) throughout the mission, which resulted in discernible improvements in cardiorespiratory fitness (VO2max) and isometric strength of the thigh muscles. No correlations were found between the change in aerobic fitness or the cumulative exercise volume and the observed decline in NK-cell cytotoxicity. We conclude that a 45-day HERA mission may not accurately capture the stressful environment experienced by astronauts during spaceflight but can cause transient immune dysregulation. Additional data acquired over future HERA missions are required to fully understand the impact that prescribed exercise training may have on immune function during ICE. Development of the Guided Imagery Component An Institutional Review Board (IRB) amendment was submitted and approved to incorporate the Guided Imagery (GI) component of the study. Our study team will interview Campaign 6 crewmembers in order to develop the GI scripts that will be used in Campaign 8. The goal of the GI intervention is to increase exercise adherence and enjoyment. Direct comparisons will be made between Campaign 8 crewmembers and Campaign 7 crewmembers, who will receive the same exercise prescriptions without GI. Dr. Judith Gordon is currently scheduling interviews with the C6 crewmembers and expects to complete all interviews by the end of summer 2024.

Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Zúñiga TM, Baker FL, Smith KA, Batatinha H, Lau B, Burgess SC, Gustafson MP, Katsanis E, Simpson RJ. "Clonal kinetics and single-cell transcriptional profiles of T cells mobilized to blood by acute exercise." Med Sci Sports Exerc. 2023 Jun 1;55(6):991-1002. http://dx.doi.org/10.1249/MSS.0000000000003130 ; PMID: 36719647 , Jun-2023
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